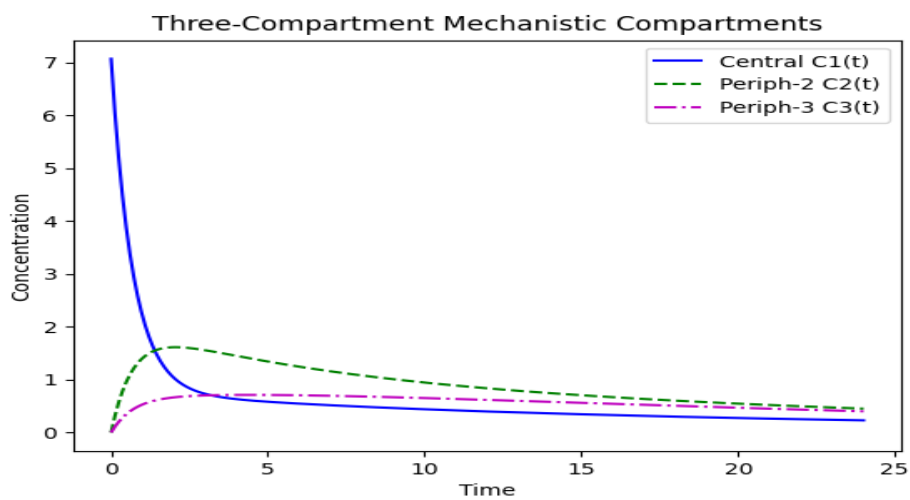


Subject S1

Central vs. Peripheral Compartments



Study Metadata

Field	Value
study_id	UserStudy1
species	Mus musculus
route	IV bolus
dose	100
model	three
time_units	h
conc_units	a.u.
allow_demote	True
k10	0.35
k12	0.6
k21	0.25
k13	0.35
k31	0.12
V1	0.04
V2	0.06
V3	0.1

subject	S1
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Data Summary

n	42
Time span (min → max)	0.00 → 24.00 h
C _{max} (observed)	7.53 a.u.
Global t _½ (log-linear)	6.13 h
m _{early} / m _{late} (log-slopes)	4.67

Fit Results (with 95% CI)

Parameter	Estimate	95% CI
A	0.869	[0.467, 1.52]
alpha	0.0437	[0.0247, 0.0761]
B	2.31	[1.83, 2.84]
beta	0.243	[0.152, 0.383]
C	3.89	[3.08, 4.66]
gamma	1.46	[0.981, 2.03]
Vd _{ss}	0.2	
Cl	0.014	

Goodness-of-Fit

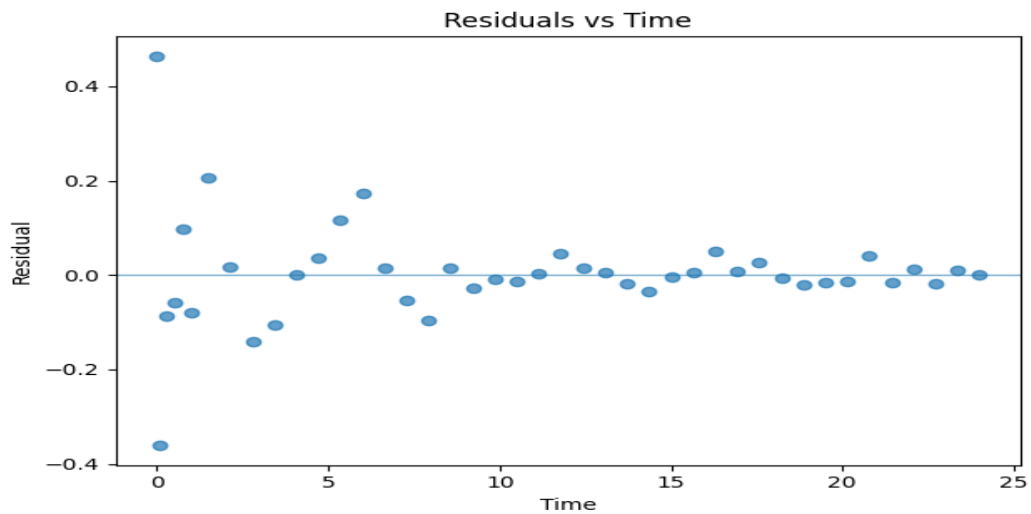
R ²	0.997
AIC	-235.7
AICc	-233.3

Model Selection Diagnostics

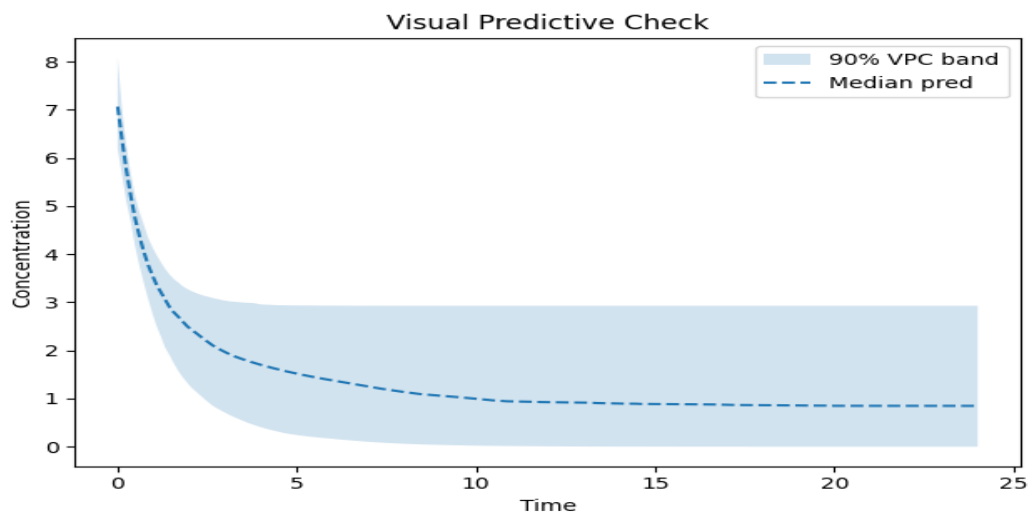
Criterion	AICc (linear)
AICc (2c)	-162.46
AICc (3c)	-170.36
ΔAICc (3c – 2c)	-7.90

rate_sep_rel (3c)	0.137
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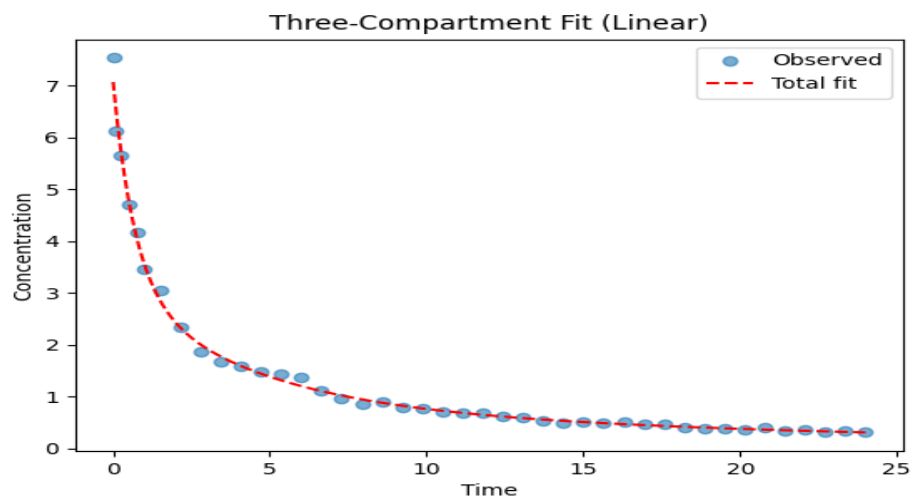
Residual Diagnostics



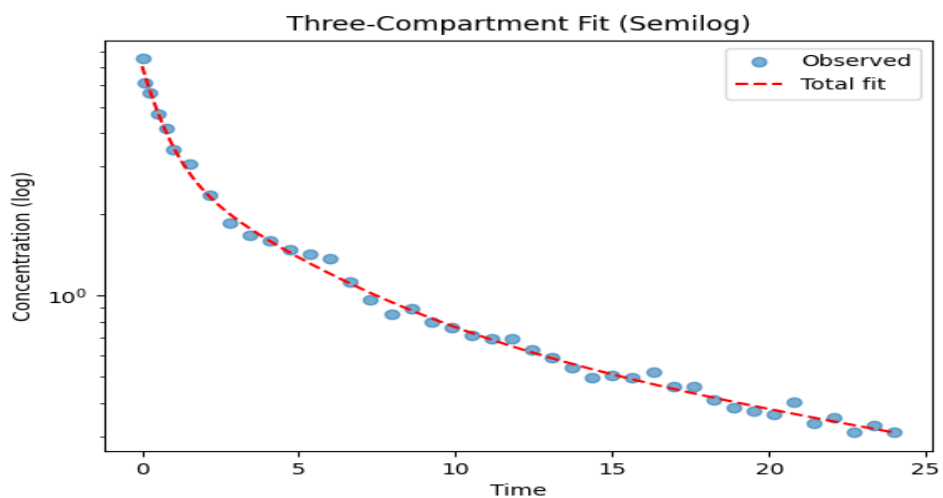
Visual Predictive Check (5–95%)



Linear Fit



Semilog Fit



Dosing Timeline

